Code-Layout, Readability and Reusability

Date	2 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability

This document evaluates the quality of the codebase in terms of its structure, readability and reusability. The project follows modular coding practices, meaningful naming conventions, proper comments and reusable components to improve maintainability.

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code.	Yes	PEP-8 formatting followed.
2	Proper File Structure	Files and folders are logically organized.	Yes	Modular project structure.
3	Meaningful Variable Names	Variables reflect their purpose clearly.	Yes	Descriptive naming.

4	Function / Method Names	Functions are descriptively named.	Yes	Readable APIs.
5	Code Comments	Inline and block comments explain logic.	Yes	Important modules documented.
6	Modular Design	Reusable functions/modules.	Yes	Separated into src modules.
7	No Redundant Code	Duplicate or unused code removed.	Yes	Refactored.
8	Error Handling	Exceptions handled gracefully.	Yes	Try/Except implemented.

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	SpeechToText	Python	Transcription Module	High
2	SemanticAnalyzer	Python	Evaluation Module	High
3	AudioFeatureExtractor	Python	Audio Analysis	High
4	PDFReportGenerator	Python	Reports	Medium
5	Dashboard UI	Streamlit	Main Interface	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well organized project structure.
Readability	5	Clear naming and comments.
Reusability	5	Reusable modules across the project.
Documentation / Comments	5	Good inline documentation.
Overall Score	**5/5**	Production-ready coding practices.